

INDEX NUMBER.....DATE.....

METHODIST UNIVERSITY COLLEGE GHANA



FACULTY OF SCIENCE

DEPARTMENT OF NURSING & APPLIED SCIENCE

(DANSOMAN & TEMA COMPUS)

END OF FIRST SEMESTER EXAMINATION, 2024/2025

B.S.c REGISTERED GENERAL NURSING

COURSE CODE: MNUR 307

COURSE TITLE: PHARMACOLOGY AND THERAPEUTIC I

TIME: 2 Hours

INSTRUCTIONS:

1. DO NOT OPEN THE BOOKLET UNTIL YOU ARE TOLD TO DO SO
2. READ THE QUESTIONS CAREFULLY AND PROVIDE ONLY RELEVANT ANSWERS
3. WRITE CLEARLY IN THE SPACES PROVIDED OR CIRCLE APPROPRIATELY USING A BLUE PEN
4. STOP WRITING AS SOON AS YOU ARE INSTRUCTED TO DO SO
5. REMAIN IN YOUR SEAT UNTIL THE PAPERS ARE COLLECTED
6. ANSWER SECTION B IN ANSWER BOOKLET

SECTION A

1. A nurse reviews the physician's prescription for a client with heart failure. The nurse expects to note which of the following?
 - A. Cardizem
 - B. Digoxin
 - C. Propranolol
 - D. Metronidazole
2. Intramuscular injection is administered at:
 - A. 80degrees
 - B. 60degrees
 - C. 90degrees
 - D. 45degrees
3. Which of the following foods would the nurse most likely recommend the patient include in the daily diet to prevent hypokalemia?
 - A. Apples
 - B. Bananas
 - C. Corn
 - D. Green beans
4. After serving a sublingual nitroglycerin, the nurse must ensure that client has which of the following most important items within easy reach.....
 - A. Call bell
 - B. Moist cloth
 - C. Tissue
 - D. Vomit bowl
5. When administering the nitrates for angina pectoris, the nurse monitors the patient for the most common adverse reaction, which is:
 - A. Anorexia
 - B. Headache
 - C. Hyperglycemia
 - D. Fever
6. The safest narcotic choice for an elderly patient with acute pain is:
 - A. Fentanyl transdermal patch
 - B. Meperidone
 - C. Morphine sulphate
 - D. Oxycodone
7. Antitussives are given with caution to patients with.....
 - A. An unproductive cough
 - B. A chronic cough
 - C. Hypertension
 - D. Hypotension

8. Which of the following serum digoxin levels would be most indicative that a patient taking digoxin may be experiencing toxicity?

- A. 0.5 ng/mL
- B. 0.8 ng/mL
- C. 1.0 ng/mL
- D. 2.0 ng/mL

9. Which of the following is a beta-blocker?

- A. Heparin
- B. Glyceryltrinitrate
- C. Propranolol
- D. Nifedipine

10. Folic acid is contraindicated in which of the following conditions?

- A. Hypothyroidism
- B. Hyperthyroidism
- C. Pernicious anemia
- D. Pregnancy

11. To avoid symptoms associated with orthostatic hypotension, the nurse advises the patient to:

- A. Avoid sitting for prolonged periods
- B. Change position slowly
- C. Get up from a sitting position quickly
- D. Sleep in a side-lying position

12. Which is the most common type of anemia?

- A. Folic acid anemia
- B. Iron deficiency anemia
- C. Megaloblastic anemia
- D. Pernicious anemia

13. When administering spironolactone (Aldactone), the nurse monitors the patient closely for which of the following electrolyte imbalances?

- A. Hypernatremia
- B. Hyponatremia
- C. Hyperkalemia
- D. Hypokalemia

14. The first stage of anaesthesia is known as

- A. Analgesia
- B. Excitement
- C. The peak
- D. Surgical anaesthesia

15. When administering an antihypertensive drug for a hypertensive emergency, the nurse:

- A. Weighs the patient before administering the drug
- B. Places the patient in a supine position
- C. Darkens the room to decrease stimuli
- D. Places the patient in a high Fowler's position

16. The peripheral vasodilating drugs are contraindicated in patients:
- A. With arthritis
 - B. With hypertension
 - C. With elevated blood cholesterol levels
 - D. During the immediate postpartum period
17. Which of the following analgesics would NOT be administered to a patient after a closure of perforated duodenal ulcer?
- A. Pethidine
 - B. Paracetamol
 - C. Suppository diclofenac
 - D. Tramadol
18. The action of aspirin is to
- A. Decrease platelet aggregation
 - B. Dilate coronary and peripheral vessels
 - C. Decrease heart rate with increase formation of thrombosis
 - D. Resists the conduction of neurons activities
19. In which of the following situations would the nurse withhold a dosage of digoxin and notify the Doctor?
- A. A pulse rate greater than 100 bpm
 - B. A pulse rate less than 100 bpm
 - C. A pulse rate of 60 bpm
 - D. A pulse rate of 72 bpm
20. A patient has been put on Lisinopril, the nurse advice the patient to look out for which common side effect?
- A. Nausea
 - B. Dry cough
 - C. Headache
 - D. Hypotension
21. If bleeding is noted while a patient is receiving a thrombolytic drug, the patient may receive:
- A. A diuretic
 - B. Heparin
 - C. Protamine sulfate
 - D. Whole blood or fresh, frozen plasma
22. Which of the following drug is derived from animal products?
- A. Atropine
 - B. Halothane
 - C. Insulin
 - D. Opium
23. When administering aminophylline, a xanthine derivative bronchodilating drug, the nurse monitors the patient for adverse reactions, which include.
- A. Bradycardia, bronchospasm
 - B. Hypoglycemia, hypothyroidism
 - C. Restlessness, nervousness
 - D. Somnolence, lethargy

24. A physician has diagnosed acute gastritis in a clinic patient. Which of the following medications would be contraindicated for this patient?

- A. Calcium carbonate.
- B. Clarithromycin (Biaxin)
- C. Furosemide (Lasix)
- D. Naproxen sodium (Naprosyn)

25. Menu, teenage patient is admitted to the hospital because of paracetamol overdose. Overdoses of paracetamol can precipitate life-threatening abnormalities in which of the following organs?

- A. Adrenal Glands
- B. Kidney
- C. Lungs
- D. Liver

26. Which of the following laboratory exams would the nurse expect to be ordered for a patient taking aminophylline?

- A. Alanine aminotransferase
- B. Electrolytes
- C. Serum aminophylline levels
- D. Thyroid levels

27. Which of the following is a good example of a vasodilator?.....

- A. Atenolol
- B. Hydralazine
- C. Lisinopril
- D. Nifedipine

28. Kwame is receiving analgesics after surgery. How often would it be given?

- A. As often as patient request for it
- B. According to Surgeon's prescription
- C. Every 30 minutes
- D. Whenever necessary

29. Which of these drugs is classified as an expectorant?

- A. Guaifenesin
- B. Codeine
- C. Dextromethorphan
- D. Diphenhydramine

30. A postoperative patient has been put on IV morphine. The patient's respiratory rate declines to 8bpm. Which medication would the nurse anticipate administering?

- A. Acetylcysteine
- B. Methylprednisolone
- C. Naloxone
- D. Protamine sulphate

31. Anaesthesia is given to a patient before surgery for the following reasons:

- I. Induce state of loss of sensation
- II. Induce sleep in patient
- III. Produce absence of pain
- IV. Produce muscle relaxation

- A. I, II and III only
- B. I, II and IV only
- C. I, III and IV only
- D. I, II, III and IV

32. After the first dose of an ACE inhibitor, the nurse monitors:

- A. The patient for a hypotensive crisis
- B. The vital signs every 4 hours or more often if the patient reports being dizzy
- C. The blood pressure every hour until it is stable
- D. The blood pressure every 15 to 30 minutes for at least 2 hours

33. A patient takes oxycodone 40mg PO twice daily, for the management of chronic pain. Which intervention should be added to the plan of care to minimize the gastrointestinal adverse effects?

- A. Eat foods high in protein
- B. Increase fluid and fibre intake
- C. Take an antacid with each meal
- D. Take the medication on an empty stomach

34. Which of the following drugs is an anti-platelet?.....

- A. Aspirin
- B. Lisinopril
- C. Nifedipine
- D. Propranolol

35. The patient is receiving the first dose of warfarin. Before administering the drug, the nurse:

- A. Administers a loading of heparin
- B. Checks to see that blood has been drawn for a baseline prothrombin time
- C. Has the laboratory draw blood for a serum potassium level
- D. Takes the apical pulse

36. Before a nurse serves digoxin to a client, she should ...

- A. Check the pulse rate
- B. Check the blood pressure
- C. Check the urine output
- D. Check the respiratory rate

37. The main reasoning for administering atropine sulphate preoperatively is to

- A. Accelerate the effect of anaesthesia
- B. Decrease secretion
- C. Improve cardiac contraction
- D. Prevent depression of the respiratory centre

38. Which of the following substances would decrease the absorption of oral iron?
- A. Antacids
 - B. Ascorbic acid
 - C. Levothyroxine
 - D. Vitamin B12
39. A patient receiving propranolol hydrochloride (Inderal) should be told to expect
- A. Acceleration of the heart rate after eating a heavy meal
 - B. Flushing sensations for a few minutes after taking the drugs
 - C. Dizziness with strenuous activity
 - D. Pounding of the heart for few minutes after taking the drug
40. Nurse Mensah is caring for a cancer patient receiving morphine sulphate for pain. Which of the following nursing actions is most important in the care of the patient?
- A. Monitor urine output
 - B. Monitor respiratory rate
 - C. Monitor heart rate
 - D. Monitor temperature
41. There is an increased risk for bleeding when the patient receiving heparin is also taking:
- A. Allopurinol
 - B. An NSAID
 - C. Digoxin
 - D. Furosemide
42. Calcium channel blockers act on blood vessels to
- A. Decrease the preload
 - B. Dilate coronary and peripheral vessels
 - C. Decrease the force of contraction
 - D. Increase thrombosis formation
43. Which drug route gives 100% bioavailability?
- A. Intravenous route
 - B. Enteral route
 - C. Parenteral route
 - D. Dermal
44. A patient taking a calcium channel blocker experiences orthostatic hypotension. The nurse instructs the patient with orthostatic hypotension to:
- A. Discontinue use of the calcium channel blocker until the hypotensive effects diminish
 - B. Increase the dosage of the calcium channel blocker
 - C. Make position changes slowly to minimize hypotensive effects
 - D. Remain in a supine position until the effects subside
45. A patient has been taking opioid for a long period of time. What is the nurse intervention?
- A. Encourage increased fluid and fibre diet
 - B. Monitor for GI bleeding
 - C. Teach the client to check blood pressure
 - D. Refer the patient to a drug treatment center

46. Heparin as administered:

- A. Intramuscularly
- B. Intradermal
- C. Subcutaneously
- D. Intrathecally

47. The nurse correctly administers montelukast (Singulair)

- A. Once daily in the evening
- B. Twice daily in the morning and evening
- C. Three times a day with meals
- D. Once daily in the morning

48. A patient admitted to the hospital with pneumonia asks the nurse why codeine has been prescribed because he does not have pain, the nurse's response is based on the knowledge that codeine also has which effect?

- A. Antitussive
- B. Expectorant
- C. Immunostimulant
- D. Immunosuppressant

49. Which of these drugs would be prescribed for a patient with an acetaminophen over dosage?

- A. Acetylcysteine
- B. Benzonatate
- C. Guaifenesin
- D. Dextromethorphan

50. Which agent below is most likely to cause serious respiratory depression as a potential adverse reaction?

- A. Hydrocodone (Lortab)
- B. Morphine (Duramorph)
- C. Nalmefene (Revex)
- D. Pentazocine (Talwin)

51. When discontinuing use of an antihypertensive drug, the nurse:

- A. Checks the blood pressure and pulse every 30 minutes after discontinuing the drug therapy
- B. Expects the doctor to order that the drug dosage be gradually decreased during a period of 2 to 4 days to avoid rebound hypertension
- C. Expects to taper the dosage of the drug during a period of 2 weeks to avoid a return of hypertension
- D. Monitors the blood pressure every hour for 8 hours after the drug therapy is discontinued

52. Reye's syndrome, a potentially fatal illness associated with liver failure and encephalopathy is associated with the administration of which over-the-counter (OTC) medication?

- A. Acetaminophen (Tylenol)
- B. Aspirin
- C. Brompheniramine/pseudoephedrine (Dimetapp)
- D. Ibuprofen (Motrin)

53. When the sympathomimetics are administered to older adults there is an increased risk of .
- A. Cardiovascular effects
 - B. Gastrointestinal effects
 - C. Nephrotoxic effects
 - D. Neurotoxic effects
54. Vitamin K is associated with which of the following function
- A. Anabolism of protein
 - B. Blood clotting
 - C. Respiratory gas exchange
 - D. Wound healing
55. Which drug would the nurse expect to be prescribed for a patient with digoxin toxicity?
- A. Any inotropic drug
 - B. Digoxin immune fab
 - C. Inamrinone lactate
 - D. Milrinone
56. the medication Rights include the following EXCEPT:
- A. Right patient
 - B. Right education
 - C. Right to refuse
 - D. Right ward
57. Which of the following is a common adverse reaction seen when administering an antihistamine?
- A. Blurred vision
 - B. Headache
 - C. Hypertension
 - D. Sedation
58. When teaching a patient about the use of vitamin B12 for pernicious anemia, the nurse would include which of the following statements?
- A. Take the oral form of vitamin B12 daily at bedtime on an empty stomach.
 - B. Take the oral form of vitamin B12 when you begin to feel weak or experience a headache.
 - C. You will require vitamin B12 injections monthly for life.
 - D. You will require vitamin B12 injections every 2 weeks until remission occurs.
59. A nurse administers naloxone (Narcan) to postoperative patient experiencing respiratory sedation. What undesirable effect would the nurse anticipate after giving this medication?
- A. Drowsiness
 - B. Increased pain
 - C. Nausea and vomiting
 - D. Tics and tremors
60. The nurse monitors the prothrombin time (PT) during therapy. Optimal PT for warfarin therapy is:
- A. Less than 25 seconds
 - B. More than 15 seconds
 - C. 1.8 to 2 times the control value
 - D. 1.2 to 1.5 times the control value

SECTION B

QUESTION 1

A patient has been put on morphine for two weeks on account of severe pain.

- Which medications will you advise the patient to avoid? (5 Marks)
- What will the nursing intervention be with regard to administration of morphine? (5 Marks)
- What would your nursing intervention be after giving a premedication of midazolam to the patient before surgery. (5 Marks)
- How will you educate a patient who is on antihypertensive to prevent falling from orthostatic hypotension? (5 Marks)

TOTAL MARKS 20

QUESTION 2

Warfarin (Coumadin) 10mg has been prescribed for a patient, as the nurse on duty you educated the patient to avoid foods high in vitamin K:

- Mention 7 foods that are high in vitamin K that should be avoided. (7 Marks)
- What education would you give to ensure the patient on discharge? (10 Marks)
- Mention three (3) adverse effects of warfarin. (3 Marks)

TOTAL MARKS 20